

Expected Value of a Random Variable

- $E[X] = \sum_{a \in S} X(a) \cdot P(\{a\}) = \sum_{v \in \text{supp}(X)} v \cdot P(X=v)$
 - $\text{supp}(X) = \text{support}(X) =$ all values with a positive probability
- Linearity of $E(X)$:
 - $E[X+Y] = E[X] + E[Y]$ if X, Y are random variables
 - $E[cX] = c \cdot E[X]$ if c is a constant
- $E[\text{Binomial}(n, p)] = \sum_{r=0}^n r \cdot \binom{n}{r} p^r (1-p)^{n-r} \leftarrow \text{way too hard}$
 so... $E[\text{Binomial}] = E[X_1 + X_2 + \dots + X_n]$, where X_i is Bernoulli(p)
 $\rightarrow E[X_1 + X_2 + \dots] = E[X_1] + E[X_2] + \dots + E[X_n] = p + p + \dots + p = np$
- $E[X]$ of geometric(p) $\rightarrow E[X] = \sum_{k=1}^{\infty} k(1-p)^{k-1} p$
 or... $X = X_1 + X_2 + \dots$, where X_i is Bernoulli($(1-p)^{i-1}$)
 $E[X] = \sum_{i=0}^{\infty} (1-p)^i$
- $E[\text{hypergeometric}(N, K, n)] = \sum_{i=1}^n X_i$, where $X_i = \text{Bernoulli}(K/N)$
- $E[\text{poisson}(\lambda)]$

5 key distributions

Summary	PMF	$E[X]$
Bernoulli(p)	$p(x=1)=p, p(x=0)=1-p$	p
Binomial(n, p)	$p(x=k) = \binom{n}{k} p^k (1-p)^{n-k}$	np
geometric(p)	$p(x=k) = (1-p)^{k-1} p$	$1/p$
hypergeometric(N, K, n)	$p(x=r) = \frac{\binom{n}{r} \binom{N-n}{n-r}}{\binom{N}{n}}$	$n \cdot K/N$
poisson(λ)	$p(x=k) = \frac{\lambda^k}{k!} e^{-\lambda}$	λ

used when drawing without replacement from a finite population
 with a fixed number of successes & failures
 - N = total population, n = num draws, K = num successes